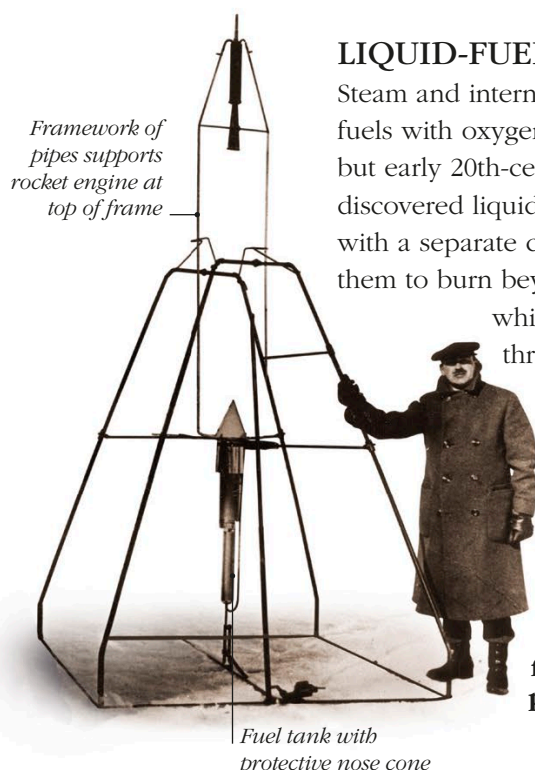


Rockets

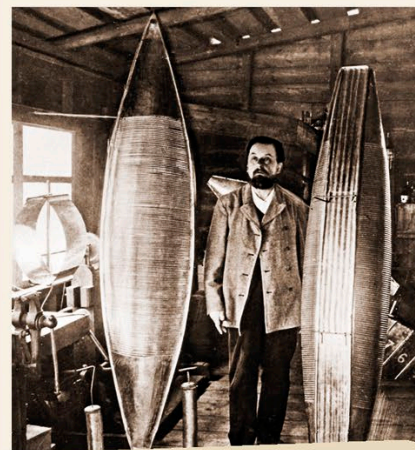
A rocket is a propulsion device that works through the principle of action and reaction. It burns fuel to create expanding gases. As the gases escape at one end, the rest of the rocket is thrust forward in the opposite direction. First used as fireworks in China in the 13th century, rockets now produce enough thrust to launch satellites and spacecraft into orbit high above Earth.



LIQUID-FUELED ROCKET

Steam and internal combustion engines burn fuels with oxygen from Earth's atmosphere, but early 20th-century rocket pioneers discovered liquid fuels that could be mixed with a separate chemical "oxidizer," allowing them to burn beyond Earth's atmosphere while also generating far more thrust. The American engineer Robert Goddard tested the first liquid-fueled rocket in 1926.

Robert Goddard at the first flight of his liquid propellant rocket, 1926

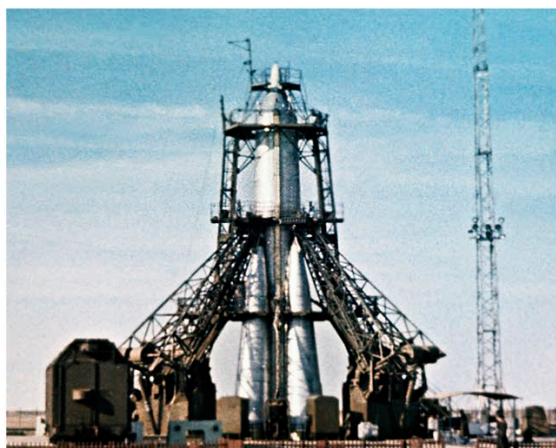


PIONEER OF ROCKETRY

The Russian schoolteacher Konstantin Tsiolkovsky laid much of the groundwork for 20th-century rocketry in a series of books and scientific articles. He was the first to suggest using liquid fuel for increased thrust, and building rockets in stages. Although he built models to demonstrate his ideas, he was never able to put them into practice.

WOW!

The first object launched into space by a rocket was a small missile fired from the top of an adapted V-2 rocket by American engineers in 1949.



SPACE RACE

After World War II, Russia and the United States raced to develop powerful ballistic missiles—rockets that could carry weapons to high altitude and then drop back to Earth, delivering weapons across thousands of kilometers. Space scientists on both sides realized that missiles could also be used peacefully to launch satellites. The launch of the first satellite, Sputnik 1, on a converted Russian R-7 missile named *Semyorka* in 1957 began the "space race" that lasted almost two decades.

R-7 rocket carrying Sputnik 1, 1957